torch.fft.rfft#

https://pytorch.org/docs/stable/generated/torch.fft.rfft.html

Description:

Computes the one dimensional Fourier transform of real-valued input. The FFT of a real signal is Hermitian-symmetric, X[i] = conj(X[-i]) so the output contains only the positive frequencies below the Nyquist frequency. To compute the full output, use fft() Supports torch.half on CUDA with GPU Architecture SM53 or greater. However it only supports powers of 2 signal length in every transformed dimension. input (Tensor) – the real input tensor

Function Signature

```
torch.fft.rfft(input, n=None, dim=-1, norm=None, *,
out=None) → Tensor#

Parameters

Keyword Arguments
```

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> t = torch.arange(4)
>>> t
tensor([0, 1, 2, 3])
>>> torch.fft.rfft(t)
tensor([ 6.+0.j, -2.+2.j, -2.+0.j])
```

```
>>> torch.fft.fft(t)
tensor([ 6.+0.j, -2.+2.j, -2.+0.j, -2.-2.j])
```

Notes & Warnings

NOTE Note Supports torch.half on CUDA with GPU Architecture SM53 or greater. However it only supports powers of 2 signal length in every transformed dimension.

Detailed Documentation

Documentation

Computes the one dimensional Fourier transform of real-valued input.

The FFT of a real signal is Hermitian-symmetric, X[i] = conj(X[-i]) so the output contains only the positive frequencies below the Nyquist frequency. To compute the full output, use fft()

Supports torch.half on CUDA with GPU Architecture SM53 or greater. However it only supports powers of 2 signal length in every transformed dimension.

input (Tensor) – the real input tensor

n (int, optional) – Signal length. If given, the input will either be zero-padded or trimmed to this length before computing the real FFT.

dim (int, optional) – The dimension along which to take the one dimensional real FFT.

norm (str, optional) –

Normalization mode. For the forward transform (rfft()), these correspond to:

"forward" - normalize by 1/n

"backward" - no normalization

"ortho" - normalize by 1/sqrt(n) (making the FFT orthonormal)

Calling the backward transform (irfft()) with the same normalization mode will apply an

overall normalization of 1/n between the two transforms. This is required to make irfft() the exact inverse.

Default is "backward" (no normalization).

out (Tensor, optional) – the output tensor.

Compare against the full output from fft():

Notice that the symmetric element T[-1] == T[1].conj() is omitted. At the Nyquist frequency T[-2] == T[2] is it's own symmetric pair, and therefore must always be real-valued.
